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Open practices in our science and our courtrooms

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Advocates of transparency in science often point to the benefits of open practices for the scientific process. Here, we focus on a possibly underappreciated effect of standards for transparency: their influence on non-scientific decisions. As a case study, we consider the current state of probabilistic genotyping software in forensics.

Calls for open practices in science gain more assent every year. Most arguments for open practices focus on the workings of academic science itself (e.g., [1]). Open practices enable replication and make research more cumulative. They also aid reanalyses of data. Open code pipelines allow a measure of error checking and verification that is impossible otherwise. Our focus is on a benefit of openness that manifests primarily outside academic science, when scientific standards are invoked in non-scientific decision-making settings. For example, in courtroom settings, expert witnesses are summoned to evaluate evidence. In many states in the United States of America (US), the Daubert standard is applied to determine the admissibility of expert testimony [2]. Under Daubert, the reliability of an expert's claims is viewed as stemming from scientific methodology, including community practices. For example, peer review and publication are considered, in spite of limitations of peer review as a filtering mechanism [3].

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within their community, but their decisions may be taken by a court to suggest that the described procedures should be admissible, perhaps even in scenarios different from those considered in the study. For scientists, there is a profound implication: the standards scientists adopt in their work indirectly affect high-stakes decisions outside science.

Probabilistic Genotyping (PG) systems in forensics

One illustrative example is probabilistic genotyping (PG) systems, described in Box 1. PG systems are designed to compare complex biological samples found at crime scenes with the DNA of possible suspects. A common application is analyzing genetic data from mixed samples – that is, samples that include genetic material from multiple people. Whereas methods for interpreting high-quality, single-source DNA samples are fairly routine, complex samples – prevalent in forensic work – are more difficult. PG systems represent a substantial advance over previous methods for interpreting complex samples.

The authors of major PG systems have published many manuscripts assessing their systems' performance (see [4]). Though there are differences among the software systems [4,5], there is widespread agreement that PG methods can perform well in an important core set of scenarios (e.g., [6] page 80).

When it comes to individual cases, however, there is disagreement about whether the results produced by PG software should be trusted. How many contributors can be assessed accurately, and on what quality of DNA sample? What extent of biological relatedness can be handled? Can PG software handle a case not just in principle, but also under the procedures used in the specific forensics laboratory that handled it? Are there bugs in the software implementation that might have affected a result?

Though studies assessing the performance of PG systems take up some combination of these and similar questions, it is difficult to cover the features of every potential case. The adversarial design of the US legal system is supposed to bridge the gap between published evidence and the specifics of the case in question. If the prosecution brings a piece of evidence and an assessment of its weight, the defense may be tasked with challenging that evidence or its interpretation.

However, the relatively closed, opaque nature of some PG software can make challenges of PG results difficult. In the United States, forensic laboratories usually use either STRmixTM [7] or TrueAllele® [8], both of which are proprietary and closed source. There are other PG systems available (see Table S1 in [4]). Many of these are open source, such as EuroForMix [9], which is widely used in the European Union. Despite occasional calls to standardize on open-source options, STRmixTM and TrueAllele® are the standard in the United States, perhaps in part because of the technical support and features they offer, and in part because they have already been admitted into court many times.

STRmixTM and TrueAllele® have proven difficult for independent researchers and defense teams to assess. First, it can be difficult for third-party researchers to obtain full versions of STRmixTM or TrueAllele® software. Independent scientists who want to evaluate STRmixTM can purchase it – price tags for a single license can run over \$15 000, not including the costs of technical support and mandatory training for personnel – or, if they are experts consulting with a legal defense team, they may be granted time-limited access under a non-disclosure agreement [access to STRmixTM software by Defence Legal teams (<https://www.strmix.com/assets/STRmix/STRmix-PDFs/Access-to-STRmix-Software-by-Defence-Legal-teams-April-2020.pdf>)]. Once the software is obtained, automated

Box 1. Probabilistic genotyping (PG) systems

Cases in which PG systems are used differ from more standard forensic-genetic analyses, in which a high-quality evidence sample – for example, large amounts of blood from a single person – is matched to a suspect. Typically, the key question is whether a person of interest is one of the contributors to the mixed sample. In the cases for which PG systems are used, samples are often of lower quality (for example, trace DNA from the handle of a firearm) and often mixed, with DNA from several contributors [1].

PG systems assess DNA evidence in light of several sources of uncertainty. PG systems take as input a set of electropherogram data produced from PCR-amplified DNA samples. An electropherogram for a given short tandem repeat (STR) locus contains ‘peaks’ suggesting the presence and relative abundance of each possible allele in a sample. Via a nontrivial set of computational steps, PG systems then produce as output a Bayes factor, usually labeled in forensics a ‘likelihood ratio’ – or, more typically, a distribution of Bayes factors produced by a Markov chain Monte Carlo (MCMC) method – comparing hypotheses that a person of interest is included or not included in the sample. Before the development of PG systems, judgments of inclusion or exclusion were made by forensic analysts examining electropherograms according to complex sets of rules. Experts were charged with considering sources of error typical to electropherogram results (‘stutter’, ‘dropout’, and ‘dropout’ of peaks), as well as uncertainty about the number and identity of other contributors, those contributors’ proportional contributions to the mixture, and the relationship between an allele’s abundance in the mixture and peak size. Results could vary considerably from analyst to analyst, even within the same laboratory.

PG systems represent an advance over previous methods for mixture interpretation, in that a formal model can be specified, and major sources of uncertainty can be described by parameters, which can either be provided by the user or estimated. For example, users can specify rates of errors like allelic dropout, which occurs when alleles present in the sample do not appear in the electropherogram, perhaps because they fail to amplify during PCR. The combined weight of evidence can then be assessed in a principled way. Though there are concerns that PG systems’ users might treat them as something of a ‘black box’ (e.g., [5]), they are less subjective than the assessments of a human electropherogram reader.

testing over thousands of samples is crucial for independent testing and validation [5]. To our knowledge, neither program offers a command-line interface (CLI) or application programming interface (API) that would make such testing practical.

Further, the source code is difficult to access. US-based defense expert witnesses who want to examine STRmix™’s source code need to travel to Ohio and view the code while being monitored [access to STRmix™ Software by Defence Legal teams (<https://www.strmix.com/assets/STRmix/STRmix-PDFs/Access-to-STRmix-Software-by-Defence-Legal-teams-April-2020.pdf>)]. It is difficult to perform serious evaluation of the code – evaluation that is best performed in conversation with extensive testing – in such an environment. Requests to review TrueAllele®’s code have been resisted, with the rationale that it contains trade secrets [10]. After denials in many cases, three judges have recently granted code review permission under protective orders. The reviews have not yet occurred. In one of the three cases,

prosecutors chose to withdraw their TrueAllele® evidence rather than undergo review [11].

In sum, it is difficult for experts who are not developers of STRmix™ or TrueAllele® to test their limits. Nonetheless, results from these systems regularly act as central pieces of evidence in decisions about guilt or innocence, including in capital cases.

In the case of STRmix™ and TrueAllele®, our most fundamental concern is not the quality or quantity of the validation studies that the authors and other scientists have completed. Rather, it is whether such assessments can be sufficient if the software and underlying code are not widely available for testing. It is good to have peer-reviewed studies in the literature supporting any method to be used in court, but peer review alone will catch only some errors and does not ensure the scientific literature is entirely correct [3]. Further, peer review of an algorithm, imperfect itself, is distinct from validation of an algorithm’s implementation in

software [12]. It is not a substitute for robust, critical testing.

There is precedent for independent examination leading to discovery of errors in forensic technology, including in at least one PG program [5] and in simpler technologies such as breathalyzers.

What can scientists do?

Beyond supporting measures that would encourage transparency in the courtroom, such as Congressman Mark Takano’s Justice in Forensic Algorithms Act, further reflection on our scientific practices may be warranted. As a scientific community, the proverbial beam is not fully dislodged from our own eye. The way in which proprietary PG systems like STRmix™ and TrueAllele® have been evaluated in the peer-reviewed literature is consistent with recent and ongoing practices in several fields. Although major strides have been made in the last decade, many journals still do not require transparency with regard to code, software, or data, even in the absence of compelling privacy rationales. Materials that are in principle available upon request may in some cases be withheld by authors without penalty.

There are many reasons offered for hesitation about blanket adoption of specific transparent practices. Some of these are compelling, especially those that concern sharing of human data, like the imperatives to protect research participants’ privacy, to heed the limited scope of participant consent, and to respect data sovereignty [13]. However, even in such cases, it is often possible to develop tailored data-sharing agreements that are agreeable to all parties.

For their part, advocates of dominant PG systems can also point to important differences between their contexts and much scientific work. For example, scientific software is usually made available without guarantees of maintenance or technical

Box 2. Research by for-profit entities

The considerations we raise here may be relevant for considering other research performed by for-profit entities. For example, when Celera published its draft of the Human Genome, an individualized data-sharing process was negotiated with the journal, leading to a set of procedures by which researchers could gain either partial or complete data access. More recent examples from human population genetics include papers from researchers at AncestryDNA or 23andMe in which some proprietary aspects of the algorithms used for data analysis are either incompletely described, or code is not released. Such papers often bring substantial benefits to the scientific community, including results of wide interest from remarkable datasets.

At the same time, there are hidden costs. Most relevant to the argument here, publishing work that relies on code that is in principle unavailable may make it easier, via *Daubert's* invocation of peer review and publication, for opaque practices to pass muster in court. The salient fact that the stakes differ – that is, that 23andMe's implementation of their ancestry inference algorithm is not going to decide whether anyone is innocent or guilty – may not be appreciated in court.

support. PG developers have adopted a closed-source, proprietary model in part to fund their efforts to provide support [14]. (Other research by for-profit entities in genetics raises related questions, see Box 2.) We emphasize that PG developers bring considerable expertise to address important problems and deserve to be compensated. But there are other models, in which private enterprises earn money by providing support for, or enterprise versions of, software that remains open source. Further, governments that rely on PG systems could fund efforts to develop and support open PG software, as the European Union supports maintenance of EuroForMix.

Our intention is not to take a position on the degree of transparency that should be required in every setting, but rather to highlight the ripple effects of our decisions regarding open practices. It is important to consider not just the contexts in which we typically work, but also the example we set as a community. Even when we pursue

questions of narrow interest and aim only to convince our close colleagues, our practices may be used as a model in other domains. A given set of practices may work well for a conversation of specialized interest, but may fail when applied in a high-stakes setting. Taking this idea one step further, we might ask how we want high-stakes decisions to proceed, and consider adopting those processes ourselves.

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